Byeongjee Kang

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Research Interests

I develop **programming language and compiler techniques** for safe, efficient execution across heterogeneous systems. My experience spans from energy-intermittent microcontrollers to SIMD/multicore servers. I leverage **MLIR** for compiler pipelines, **WebAssembly** for heterogeneous deployment, and **Rust** for developer-facing interfaces.

Education

Carnegie Mellon University — Pittsburgh, USA
Ph.D. in Software Engineering, advised by Dr. Limin Jia — *Aug 2023 – Present*

Pohang University of Science and Technology (POSTECH) — Pohang, Korea
Bachelor of Computer Science and Engineering & Mathematics (double major) — *Feb 2018 – Feb 2023*

GPA : 4.26/4.3 (Ranked 1st among all graduating students in class of 2023)

Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA

Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021)

Professional Skills

Programming Languages: Rust, C++, WebAssembly, SML, Python, JavaScript, Scala, Maude, Futhark

Frameworks: LLVM, MLIR, React, DJango, Git

Work Experiences

Microsoft Research — Redmond, WA, USA
Research intern at Data Systems Group advised by Dr. Kaushik Rajan — *May 2025 – Aug 2025*

Compiler for database intrinsics: Auto-vectorizing compiler on MLIR with support for control flow dependencies

Improved the performance of database intrinsic functions with while loops, which could not be vectorized by existing compilers

Kodebox — Seoul, Korea
Software engineer as military service alternative — *Jan 2020 – Dec 2021*

CodeChain Foundry: Open-source blockchain engine based on composable module system

Worked as core developer of consensus engine, implementing BLS signature aggregation scheme and designing interface for modules to communicate with consensus engine

Zuzu: Web-based captable management software

Worked as full-stack software engineer

Publications

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM
Ao Li, Byeongjee Kang, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye
Object-oriented Programming, Systems, Languages, and Applications (OOPSLA) 2025 (to appear) (pdf)

GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits
Sam Westrick, Pengyu Liu, Byeongjee Kang, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding and Umut Acar
IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (**Best Paper**) (pdf)

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems
Byeongjee Kang, Kyungmin Bae
Science of Computer Programming 2024 (pdf)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search,
Byeongjee Kang, Kyungmin Bae,
International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022 (pdf)

Preprints

WAMI: Compilation to WebAssembly through MLIR without Losing Abstraction
Byeongjee Kang, Harsh Desai, Limin Jia, Brandon Lucia, (pdf)

Mr3: An Execution Framework for Hive and Spark
Sungwoo Park, Seunggon Namgung, Byeongjee Kang, (pdf)

Talks

An MLIR Dialect for WebAssembly, WebAssembly Workshop 2025, [recording]

Ongoing Research Projects

Automatic Program Rescaling and Migration May 2025 – Present
Programming language and runtime support for automatically rescaling programs (including ML applications) for resource-constrained systems

Aiming to automatically rescale programs with correctness guarantees using general approaches such as program slicing, as well as ML-specific techniques such as quantization and sparsification

Probabilistic Modeling and Analysis Tools for Intermittent System August 2025 – Present
Probabilistic modeling of resource usage in intermittent systems

WAMI: MLIR-based Compilation Pipeline for WebAssembly October 2024 – Present
MLIR-based compilation pipeline for WebAssembly

Honors and Scholarships

ILJU Fellowship (ILJU Academy and Culture Foundation) *Aug 2023 – July 2028*
Received total of USD 120,000 over eight semesters during graduate studies

Program for Highly Dedicated Students (POSTECH) *Feb 2018 – Feb 2023*
Presidential scholarship and supporting program in POSTECH offered to **one outstanding freshman** each year

Presidential Science Scholarship (Ministry of Education of Korea) *Feb 2018 – Feb 2023*

National science scholarship offered to top freshmen students in the field of science and technology

Last updated: Oct 4, 2025